

MASTER THESIS

Immersive Authoring of Opportunistic Interfaces in Mixed Reality

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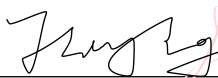
August 19, 2024



Declaration of Authorship

I, Jiaying LU, declare that this thesis titled, “Immersive Authoring of Opportunistic Interfaces in Mixed Reality” and the work presented in it are my own. I confirm that:

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- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
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“Somewhere, something incredible is waiting to be known.”

Carl Sagan

TECHNICAL UNIVERSITY OF DRESDEN

Abstract

Institute of Software and Multimedia Technology.

MSc.

Immersive Authoring of Opportunistic Interfaces in Mixed Reality

by Jiaying LU

This thesis makes use of the advancement of mixed reality headsets, enables users to turn surrounding objects into functional interactive 3d widgets and gives them the freedom to do creative authoring of the tangible widgets, which could lead to more accurate physical feedback and more customized mixed reality user experience. Those interactive 3d widgets are referred to as opportunistic interfaces in literature. However the creation of such opportunistic interfaces remains challenging. Existing works often rely on complex object tracking algorithms and even additional hardware. There is a lack of functioning systems despite the concepts in some of the current papers. This work aims to provide a pure software solution with consumer level headsets such as the Oculus Quest and explore the possibilities of constructing such opportunistic interfaces with hand tracking based immersive interactions and some other supplementary functions. This work would focus on the real feasibility and output of the idea. Moreover, this work plans to demonstrate the outcome with small scale daily objects, room scale objects, and some example situations. The achievement of a previous research project, such as spatial anchors and area of interests are used as a starting point.

Acknowledgements

Hereby I would like to express my gratitude to the support of my supervisor, my friends and my parents.

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For/Dedicated to/To my...

Chapter 1

Introduction

1.1 Background

Since the introduction of virtual reality in the 1960s, it has gone through more than sixty years of development. It was originally developed for a variety of applications in different fields, but once met the bottleneck of technological progress and commercialization because of the hardware limit twenty years ago.

Nowadays in 21 century, hardware technology has been sufficient to support the needs of VR industry, there is enough potential and possibilities for Virtual Reality technology to take a step ahead and into People's daily lives.

Traditional Virtual Reality was based on completely virtual scene composed of 3D modeling and simple haptic feedback provided by attached devices like controllers, which leads to a limited physical interaction and unnecessarily high demand of virtual scene building.

Actually, involving real world environment as a part of the virtual scene would not destroy the realism that virtual scenes bring, but even offer more potential for interaction. So there comes Mixed Reality, which is a further development of virtual reality technology, it enhances the realism of the user experience by introducing information about the real scene into the virtual environment, building an information loop of interactive feedback between the virtual world, the real world and the user.

Mixed reality is a combination of a set of technologies that not only provide new ways of viewing, but also new ways of inputting, these features would benefit each other and encourage innovation. In this way, mixed reality can have a direct impact on users' workflow, helping them to be more productive and innovative.

1.2 Motivation

With the development of Mixed Reality technology, immerse media will be used in more and more fields in the future and will serve as a powerful tool for deeper involvement in production and research in these fields.

In the process of development, haptic feedback is taking a more important place in immersive experience, which provides users with a third sense in addition to simply visual and auditory. Which means, including the real world environment into the immersive scene would be a critical part of virtual experience.

In the ideal future of immerse media application, it will be put into all walks of life, so the utilization space of Mixed Reality will not be limited to a simple empty room or a preset lab, experience with the headset could be taken place anywhere in our life: in office, park, museums, on public transportation or any other possible circumstances. Not only for the safety of the users, but also for better feedback in

different scenarios, maximizing the combination of virtual scenarios and real environments, virtual scene need to have the ability to adapt to real objects.

However, under different realistic scenarios with very different objects in reality, it will become very difficult to provide effective physical feedback for virtual scenes. Therefore, fixed predefined UI components will be difficult to apply to the changing Mixed Reality scenarios. This project is dedicated to provide designs and some demos for user-defined opportunistic UI to propose the possibility of customized UI, as well as the integration of immersive media into daily life in the future.

1.3 Project Introduction

This Project aim to build a basic system of opportunistic user interface in immersive environment with the help of Mixed Reality headsets. It enables users to turn surrounding real world objects into functional interactive 3d widgets and encourages them to do creative authoring of these tangible widgets, which could lead to more accurate physical feedback and more customized mixed reality user experience.

1.3.1 Opportunistic UI Definition

Opportunistic Controls are defined as a class of user interaction techniques that could support gesturing on, and receiving feedback from, otherwise unused affordances already present in the domain environment.[3] It supports gesture recognition, haptic feedback, interactive input and self-explaining functionality of the control.

Opportunistic user interface is designed as a connection between virtual scene and real world environment, the virtual user interface would add interactive widgets on real world environment, while the real object could offer the demanded physical feedback.

1.3.2 Former Works

Tangible widget is a concept that has existed for a long time, and its not only a part for immersive media. a lot of works has proven that the users' operation speed would increase significantly with tangible objects.[7]

And also, making use of real world objects for haptic feedback is not a brand new topic, many designs like interactive screen casting desktop could already be seen in recent years. Interactive experience is more natural for users to understand and easier to operate.

But in this field, there are still more theoretical conception and design innovation than realization, and the application of such techniques are mostly still remained in a small range like entertainment business.

1.3.3 Problems Remained

A lot of existed works are based on Augmented Reality instead of Mixed Reality or Virtual Reality, which means they focus more on augmented real environments rather than immersive experiences and mostly programmed for non-immersive platforms like laptop or mobile platforms.

Some works with the Mixed Reality headsets are based on complicated tracking systems in order to detect the surroundings, they are highly dependent on extra hardware like additional cameras or calibration stations.

There are also some detailed investigations about comfort level of using the virtual user interface and how to satisfy users' needs and usage habits, but remains a useful foundation of the concept rather than output of functioning systems.

1.3.4 Innovation

This project enables users to create, place, change and define the functionality of user interface widgets according to their own demands and preferences. In this way, they could minimize offset and ignore the error between real objects and calibrated virtual scene.

Customized user interface can be adjusted any time and receive haptic feedback regardless of the change of real world surroundings.

This project carries no dependency on any additional devices, the entire system is presented only by the Oculus Quest Headset.

Chapter 2

Opportunistic User Interface Widgets

2.1 Introduction of 3D Widgets

Widget is an encapsulation of geometry and behavior used to control or display information about application objects.[4] Three-dimensional widgets are based on two-dimensional widgets but with a lot of more possibilities, not only they could make use of as much as possible degrees of freedom, 3D widgets also present more sophisticated visual information. Instead of simple clicking or pressing, more kinds of interactions with physical feedback can be provided. And a lot of 3D widgets are simulation of real world interactive objects, which means a more familiar figure to users and more self-explaining using approach.

2.2 Command Button

Command Button is the most common widget in most of the programs. It could provide very simple but useful functions of triggering events, as well as starting, interrupting, and ending a process.

The most common event that a button receives is a Click event, and there are normally two states of a button, the original state and the pressed state. In this project, the virtual command button in immersive environment would simulate people pressing a real button in daily situation.

2.2.1 Prefab

The Configurable Button Prefab contains a 3D button widget, and a user interface panel, the button is presented in a little black plate, and it would only be activated after it is confirmed.

The user interface panel provides the functionality of confirming and canceling confirm the prefab, deleting it, and some additional information.

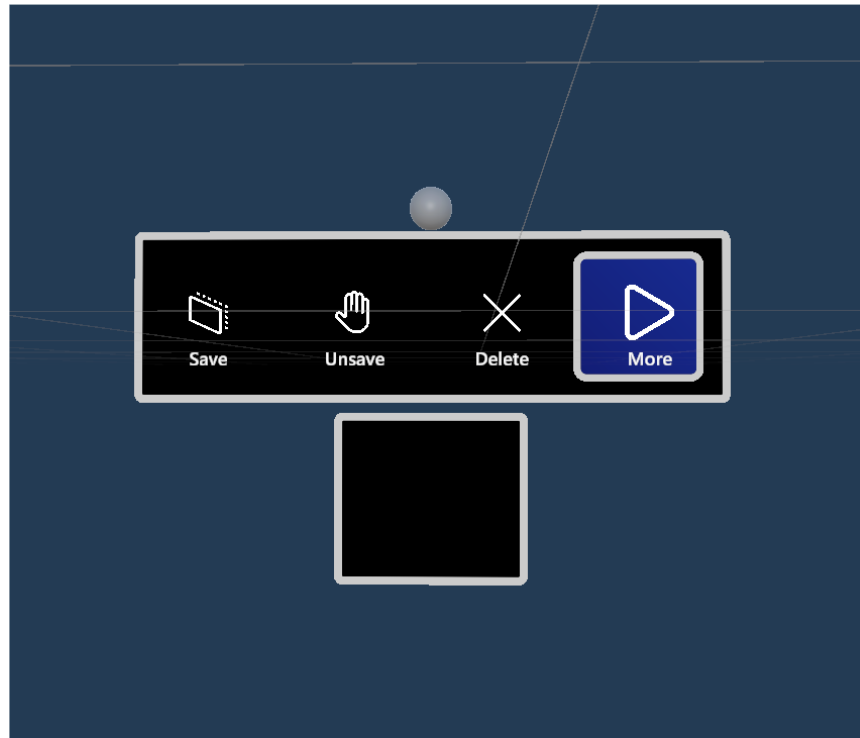


FIGURE 2.1: Button Prefab.

2.2.2 Placement

The Configurable Button Prefab is movable before it is confirmed. After generating a button prefab, users could place it wherever they want, and if the black plate of the button is aligned to a touchable real world object, users can get physical feedback when pressing the button, which is similar as using a command button in real world.

This project has came up and realized two approaches enabling the users to change the position of button widgets.

1.RigidBody

The first approach is to add a RigidBody to the prefab, so the users could push them directly with rendered virtual hands (or controllers).

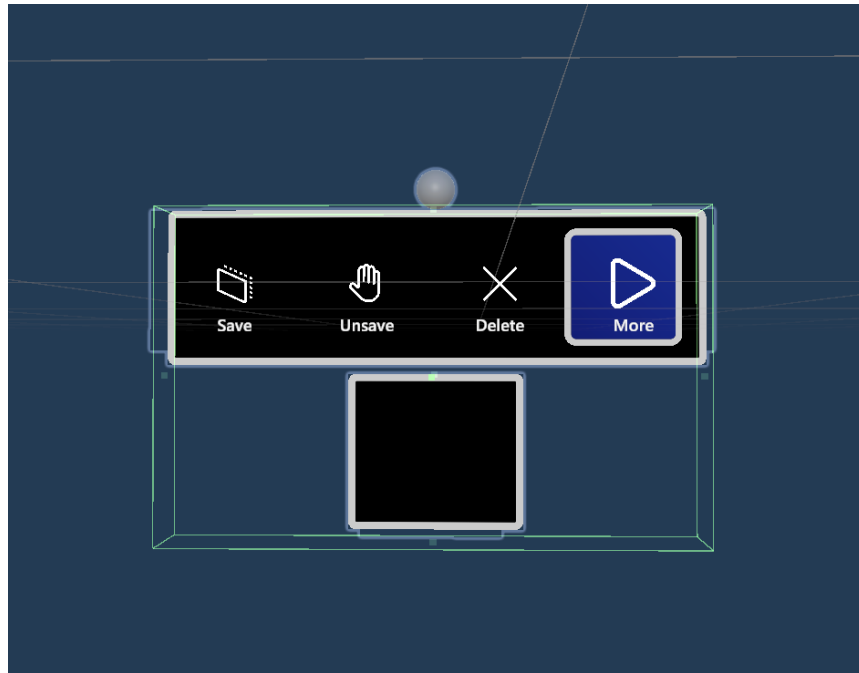


FIGURE 2.2: Box collider.

Pros: The simulation is very simple and direct, so the usage would be very intuitive, users could easily come up with how to perform this approach even without any additional explanation.

Cons: The rendering of virtual hands could not precisely represent the differences of users' hands, and the physical engine of Rigidbody is mainly designed for collisions instead of accurate alignment, so the errors might increase.

Pressing the buttons also rely on the box collider of the button itself and buttons on user interface panel, multiple box collider for parent object and child objects will overlap and cause difficulties pressing the buttons in experiment.

The Rigidbody could only react to force but not some other precise operations, if the button prefab is pushed to far and blocked by real world objects, it could be tricky to interact with.

2.Attached to fingertip

Another approach allows the user to attach the button to their left index fingertip when they pinching with right hand and close enough to the prefab. They could adjust the position and angle by simply moving their left index finger and decide it by releasing pinching.

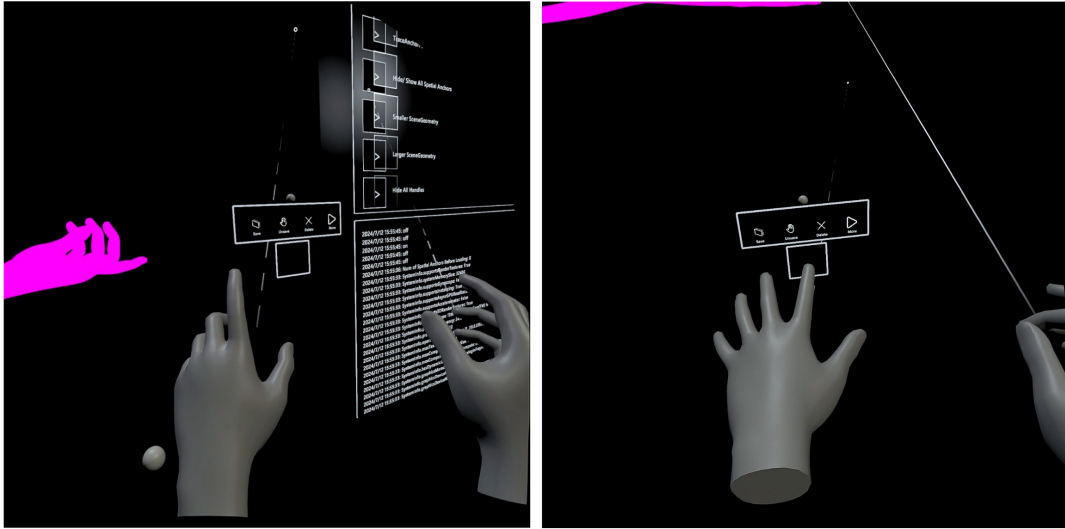


FIGURE 2.3: Pinch and Snap.

Pros: The users can see the position and rotation of the prefab directly, and its already in the same way when they are pressing the button, so the placement will be very accurate, and they could easily determine the angle by rotating their finger.

Users could get to control the prefab within a minimal distance (2-3 centimeters in most cases), so even the prefab is blocked a little by real world objects, it will not affect the usage.

Cons: The perform of this approach will need extra explanations.

The action of pinching contains too much other commands in Oculus, it might be mixed with other basic functions like clicking and dragging objects and cause difficulties in usage.

3.Comparison

Since the accuracy of bare hand interaction is based on the rendering of Oculus OVR-Hand, and the spatial calibration based on spatial anchor would always be with little offset despite all the improvements, there will be some inevitable differences between presented virtual model and real object feedback. But in either approach the users are using their hand or finger to lead the button prefab to ideal position, and they would also press the button with the same action, even there is differences between the virtual button widget and objects in the real world, it would automatically be canceled out, and they will still get the physical feedback as planned.

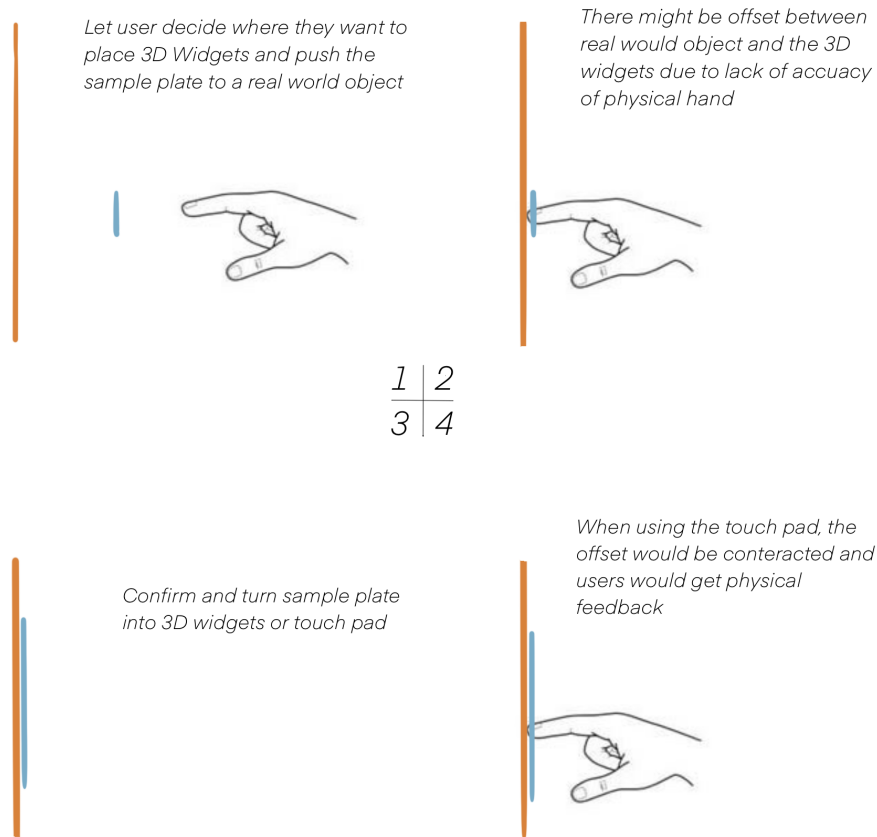


FIGURE 2.4: Offset Cancel Out.

Both approaches realized the user customized placement of button prefab, after experiment, this project will choose attaching to fingertip as the placement method for button prefab. Using Rigidbody have to take some physical factors into consideration, like gravity, inertia and forces, and changing the rotation of such a small size object could be difficult to control, which might lead to larger inaccuracies. Also adding Rigidbody to a button prefab needs to enable box collider to the whole prefab, during experiment, it turns out to cause conflict between moving the prefab and pressing the buttons.

To solve the existed problems of finger snapping, a control panel is added to the project, with which the users could declare to switch on or off the “Moving button” mode, the snapping with pinching would only be effective if it’s switched on. After defining the position of the button prefabs, users could switch it off so it would not affect other operations.

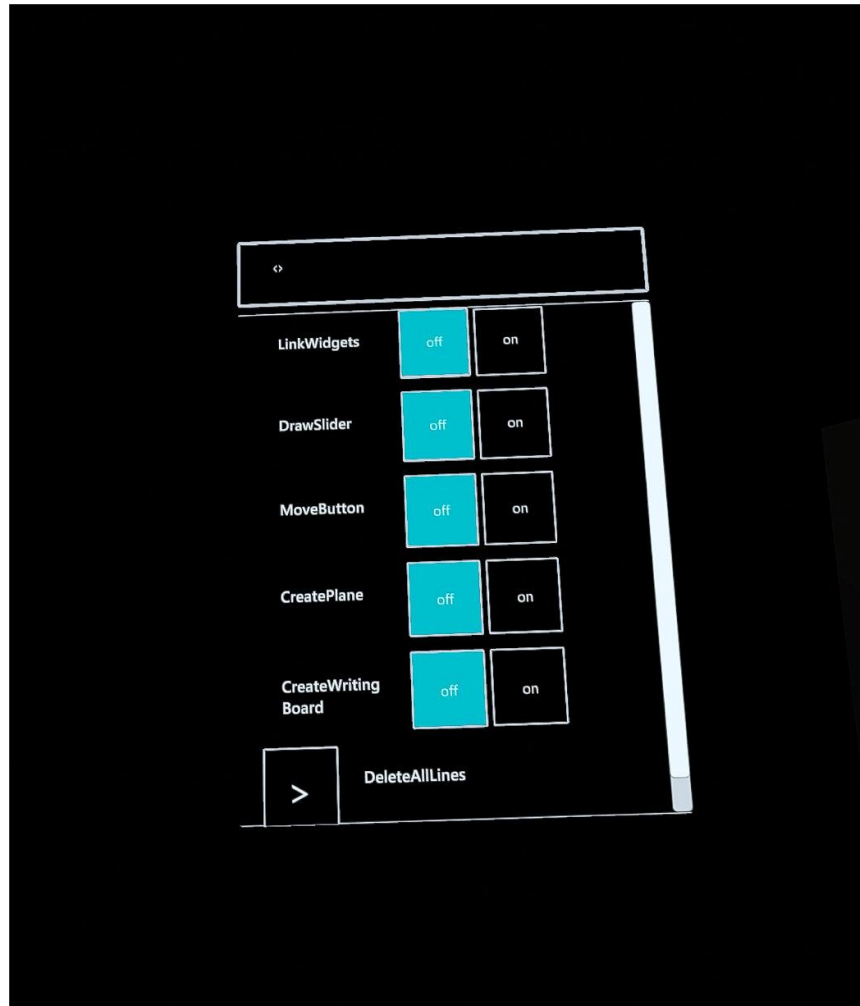


FIGURE 2.5: Control Panel.

And also, the instruction of all the demanded information will be added to the additional board of the prefab.

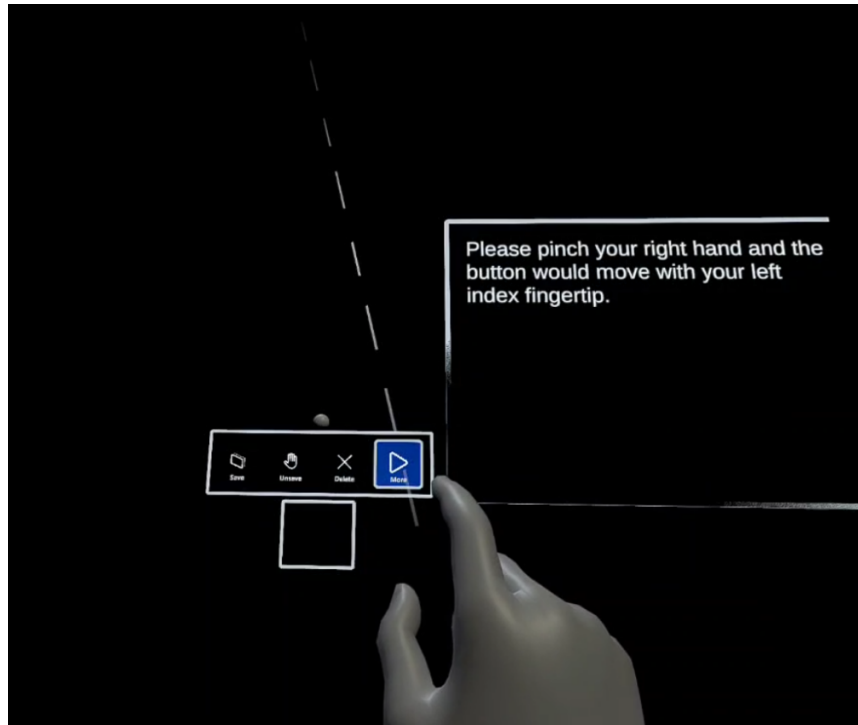


FIGURE 2.6: Additional Information Panel.

2.2.3 Generating

Users could easily create a new command button by simply clicking the 'generate button' on the control panel, which is self-explaining and easy to operate in the system, and the button would be generated in the center of the scene.

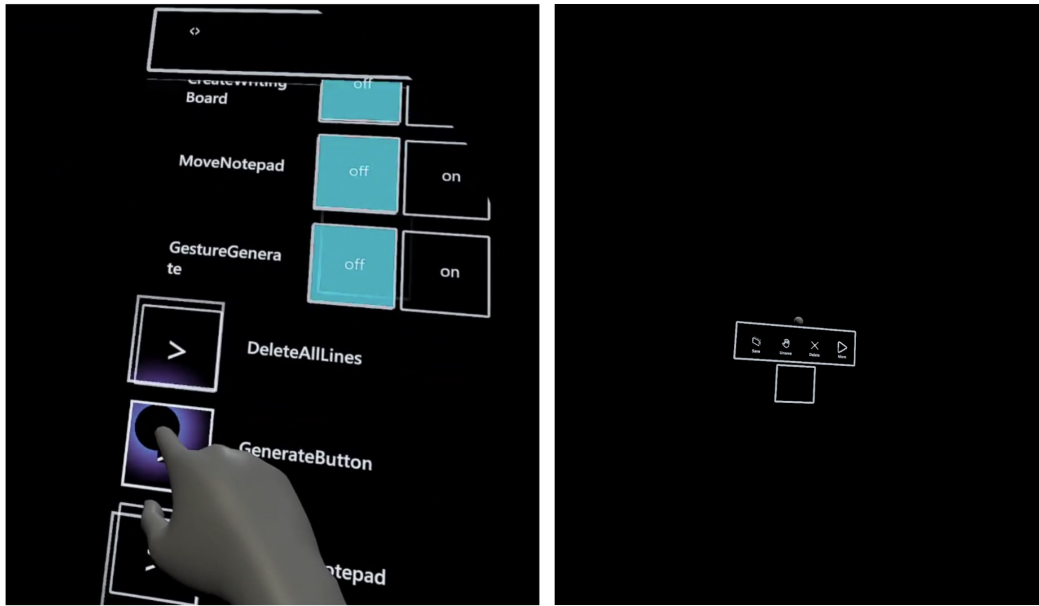


FIGURE 2.7: Command Button Generating.

2.3 Slider

Slider is also an important component of the control, it is a graphical control element with which a user may set a value (within a fixed range) by moving an indicator. Sliders are widely used in adjusting discontinuous values and controlling various targets, like representing size, numerical value, proportions, and viewing and jumping to a playback position in media player software.

In order to maximize the benefits of mixed reality and provide larger operational freedom, this project came up with two kinds of 3D sliders.

2.3.1 Straight Slider

1.prefab

Unity only provides 2D slider widgets even in 3D scenes, with has a huge limit in form, usage and interactions. So the straight slider prefab presented in this project is a combination of track stick and indicator consist of 3D game objects.

The slider prefab also contains user interface panel with the abilities of confirming and canceling confirm the prefab, deleting it, and other additional information.

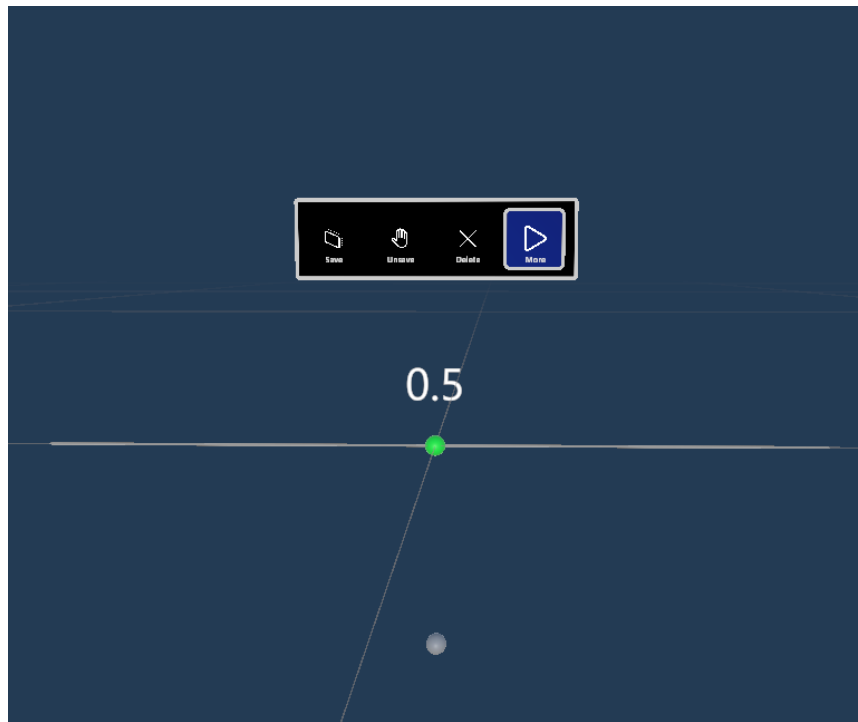


FIGURE 2.8: Straight Slider Prefab.

2. Generating

This System allows users to define the length, position and rotation of the slider, moving and placing a virtual slider precisely could be difficult, and rotating the Rigidbody of a stick-shaped object is hard to operate, so users are encouraged to define the size, position and rotation of a slider prefab while generating. In this project, they could 'paint' the progress bar of the slider by themselves.

In 3D space, it is not possible to determine when to start or end drawing like pen tip touching or leaving paper surface in the real world, so this project introduced a simple logic. After turning on the 'draw slider' mode in control panel, users could draw a sample line by moving their left index finger while pinching their right hand. The line they painted would be rendered into a functional slider when they release pinching.

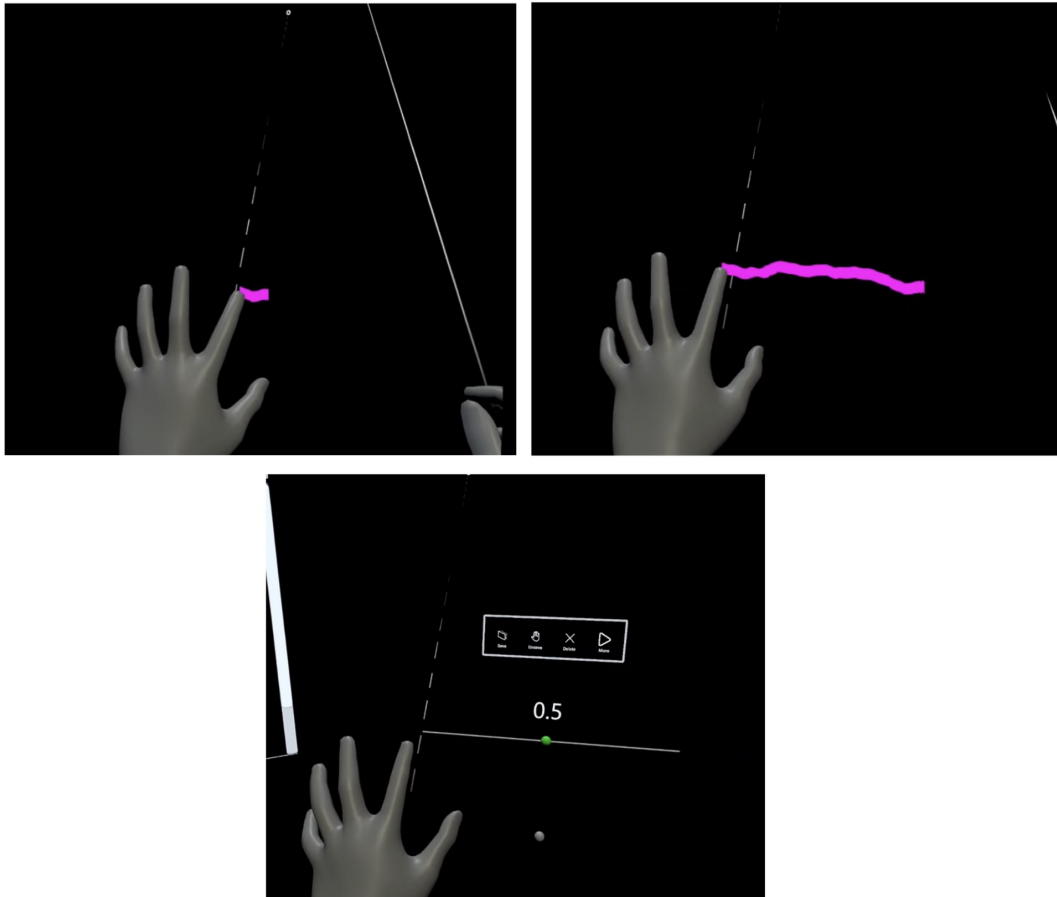


FIGURE 2.9: Generate Straight Slider.

2.3.2 Customized Slider

1.prefab

Instead of a complete slider with pointer and track, customized slider prefab only consists of the pointer and additional user interface, shape and length of slider track is one hundred percent defined by users.

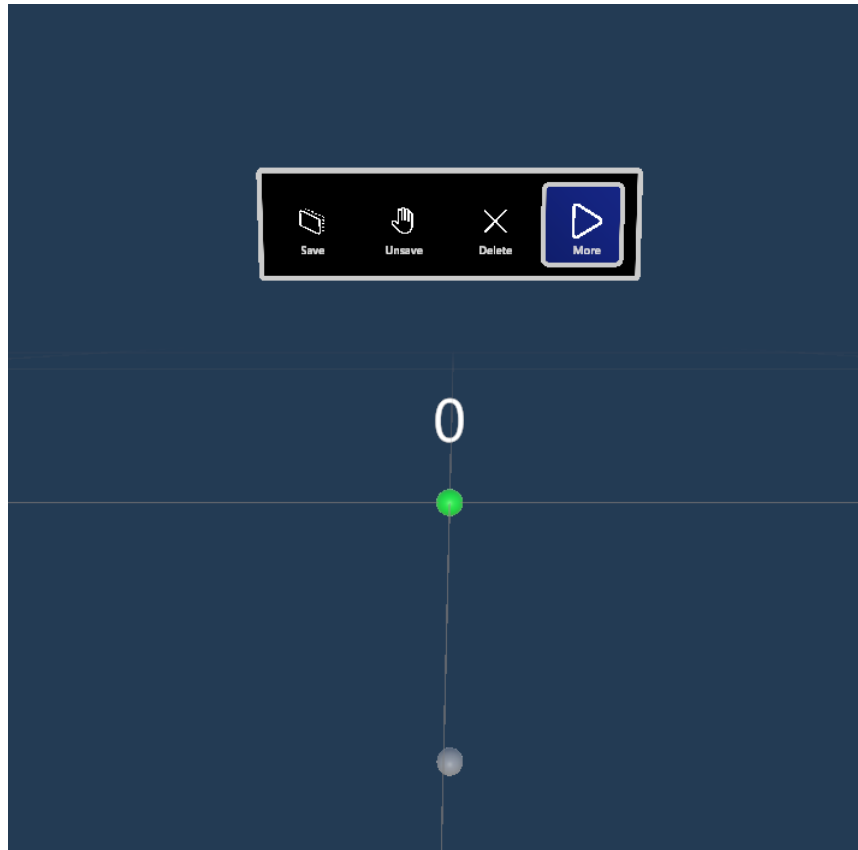


FIGURE 2.10: Customized Slider Prefab.

2. Generating

The generating process of a customized slider is also by painting while pinching after switching on the 'Customized Slider' mode on control panel. Users could draw any kind of lines in the space as the track stick, lines would be rendered into a slider.

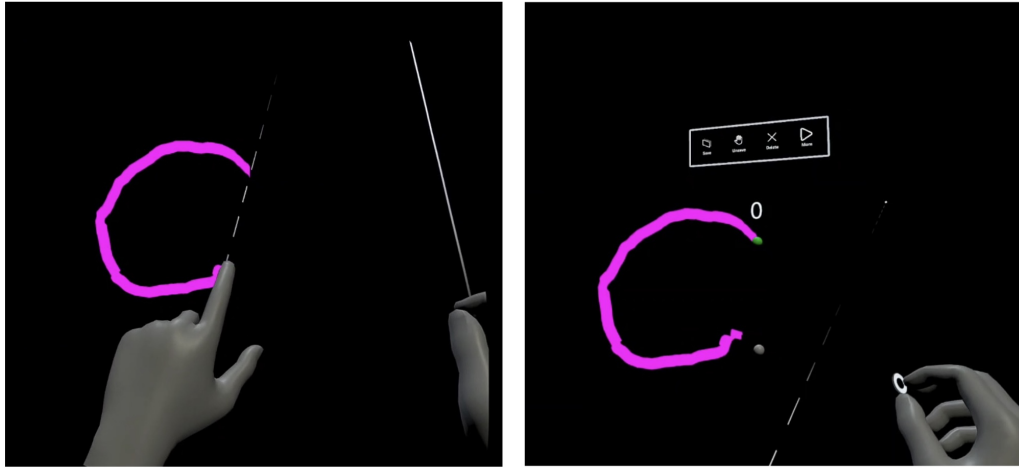


FIGURE 2.11: Generate Customized Slider.

A customized slider breaks out of its limitations in 2D space, involves three-dimensional space to be part of its generating. It could be aligned to objects with varies shapes and outlines, and could reflect the value or progress better in certain circumstance. The value of the slider will be defined by the position of pointer on the track stick.

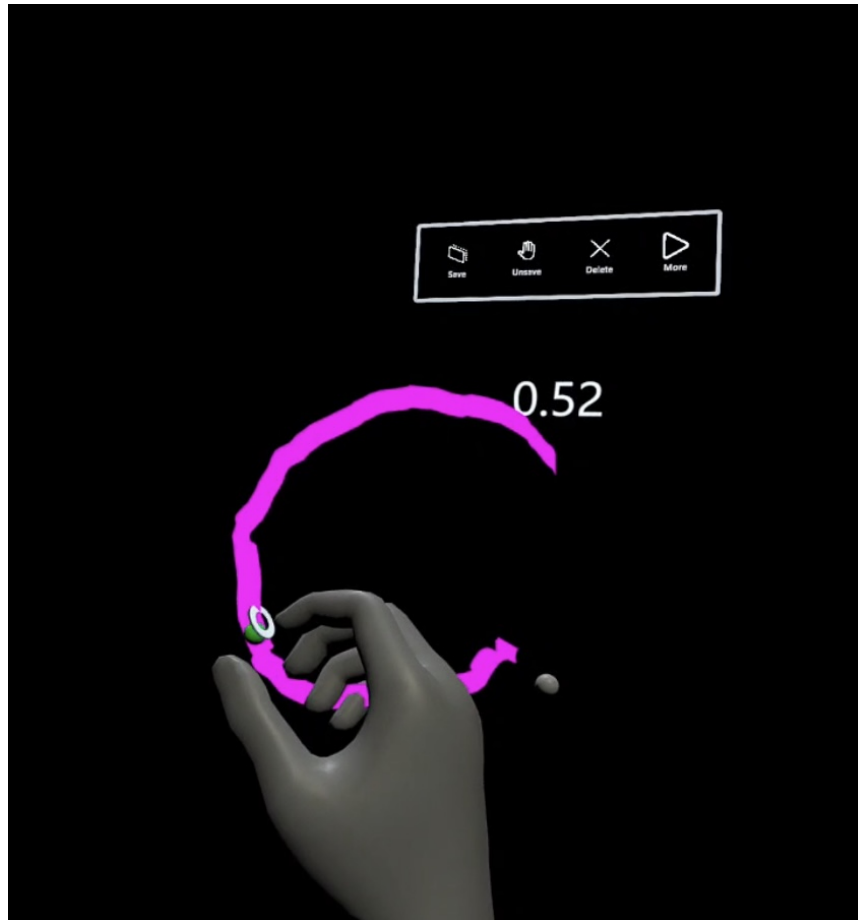


FIGURE 2.12: Customized Slider Value.

2.3.3 Comparison

On one hand, straight sliders are more in line with peoples' common sense and usage habits, and a straight progress bar is easier to read, one can easily adjust the pointer to the desired value with intuition.

On the other hand, customized sliders give more freedom and potential, users could do any kind of creative generation of the sliders. Also, by aligning to real world objects during generating, they are able to get physical feedback on any type of objects.

The existence of these two kinds of sliders don't conflict with each other, users could choose either one freely, so both prefabs are preserved in the system.

2.4 Screen

As a significant part in 2D scenes, screen also play an important role in 3D projects. Benefiting from the possibilities of interaction in three dimension environment, screen now contains more potential. The definition of a touchscreen should be a display device which also takes input via embedded pressure sensors, which would also be a key feature of the 3D touchscreen.

2.4.1 Screen Prefab

The screen prefab contains a default plane representing the size, rotation and position of the upcoming touchscreen, a linking slot for authoring, and a user interface panel with the abilities of confirming and canceling confirm the prefab, deleting it, and other additional information.

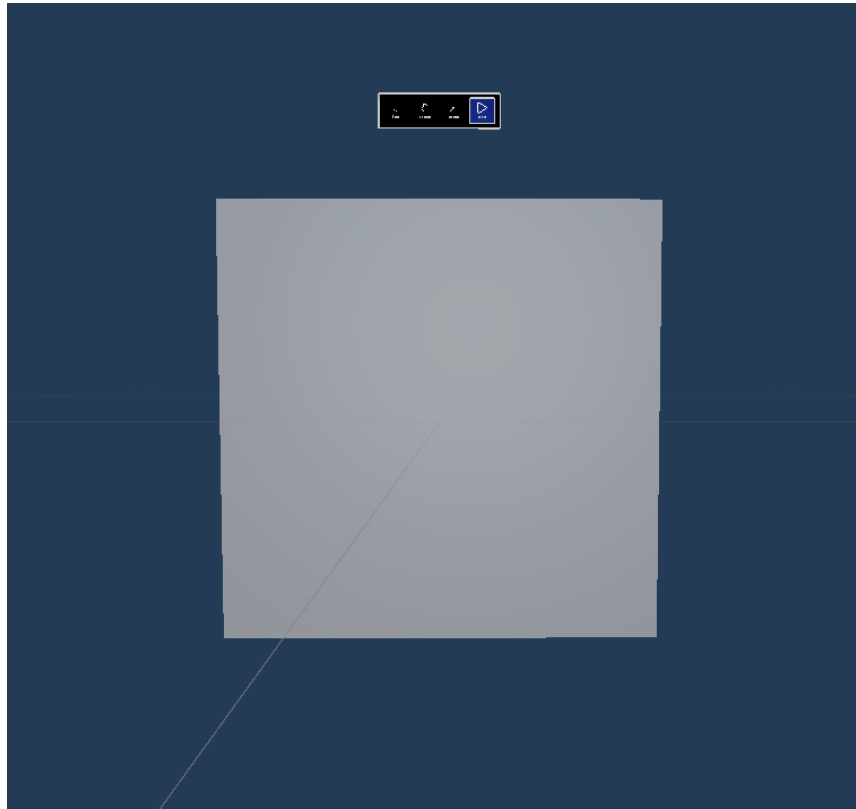


FIGURE 2.13: Screen Prefab.

2.4.2 Generating

When the “generating Screen” checkbox on the control panel is switched on, users could pitch in the air as the ideal starting point of a screen prefab (normally the upper left corner of the screen), and drag to decide the size of it, simulating the click-and-drag movement of the mouse when generating a multiple selection or painting a rectangular in 2D cases. After loosing the pinch, a screen prefab would be added to the scene, which is temporarily a default plane with no specific function.

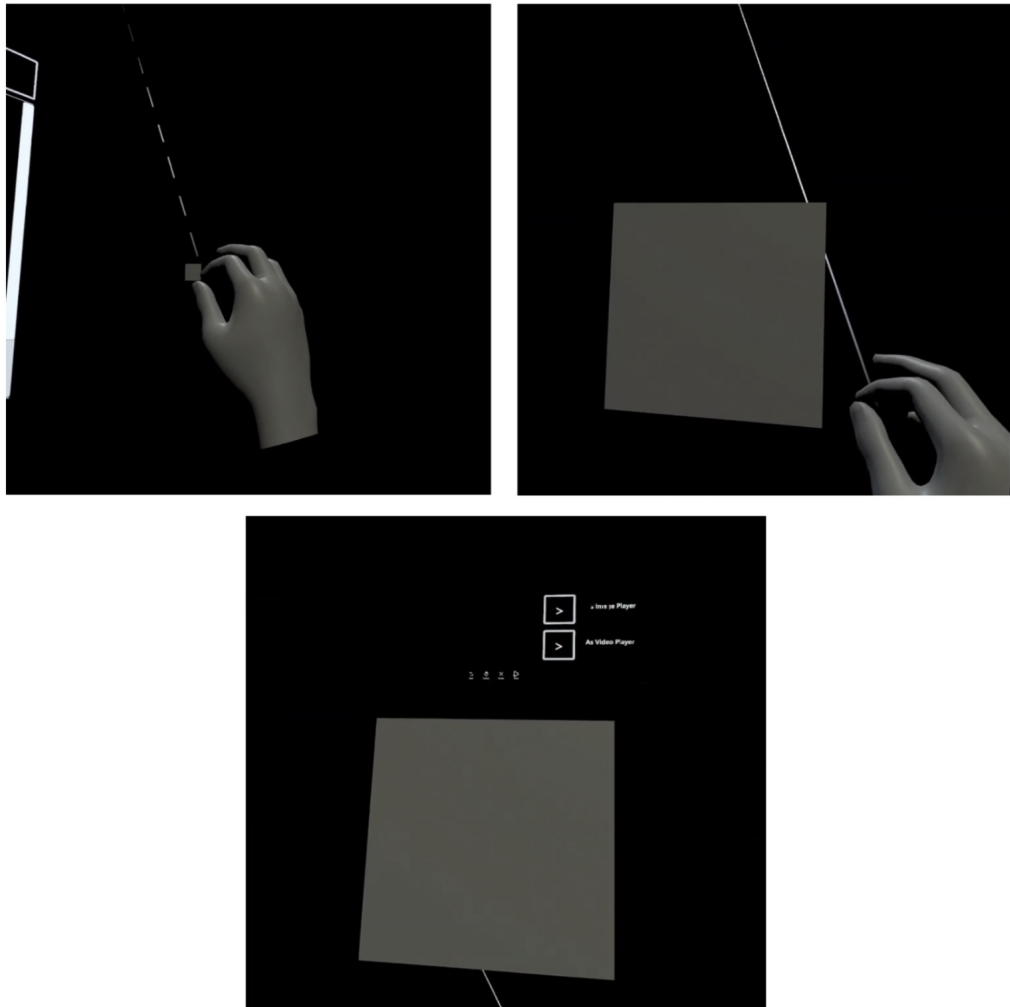


FIGURE 2.14: Generating Screen Prefab.

2.4.3 Switching to Image and Video Player

The additional information button on the UI panel contains the function enabling users to switch the plane into a screen to display images or demonstrate slides, or a video player.



FIGURE 2.15: Switch to Image Player.

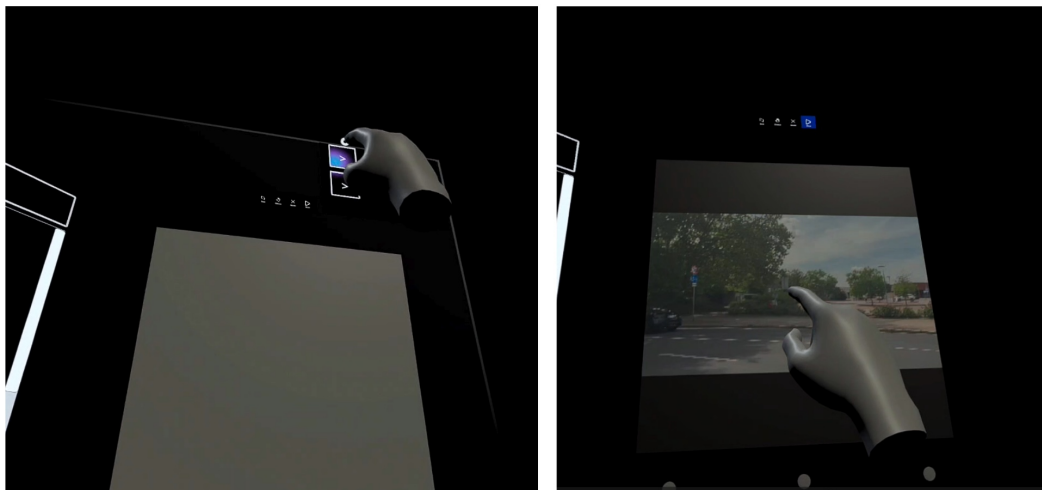


FIGURE 2.16: Switch to Video Player.

The switch could still take place again any time during using.

2.4.4 Image Player

The screen prefab could be used as a image player to show a picture, it could be a photo player, or simply a changeable sign or decoration in the immersive scene. It could also demonstrate slides as long as they have been pre-tranferred into pictures in the form of .jpg, .jpeg, or .png.

There is two invisible buttons on the screen, which only shows when the virtual hand of users are close to the screen. With pressing the button users are allowed to switch to next picture or turn the page of a slides like running a powerpoint in immersive scene.

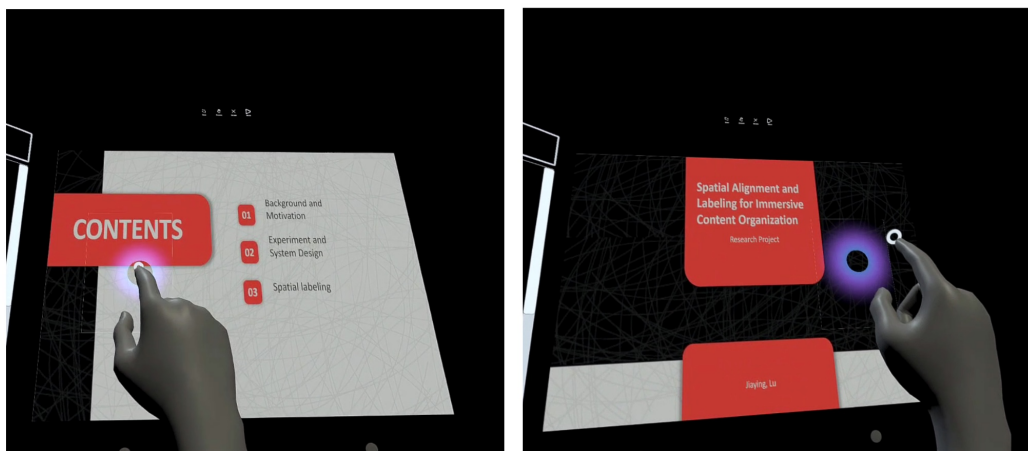


FIGURE 2.17: Image Player.

2.4.5 Video Player

The screen prefab could be turned into a video player which is functional with any kind of prepared video source. The size of the video would automatically adapts to the screen size.

There is three button on the screen only visible with virtual hands nearby. The middle one is for starting, pausing and continue playing the video. And the buttons covering each side of the screen is for forwarding or playing backward the video for fifteen seconds.

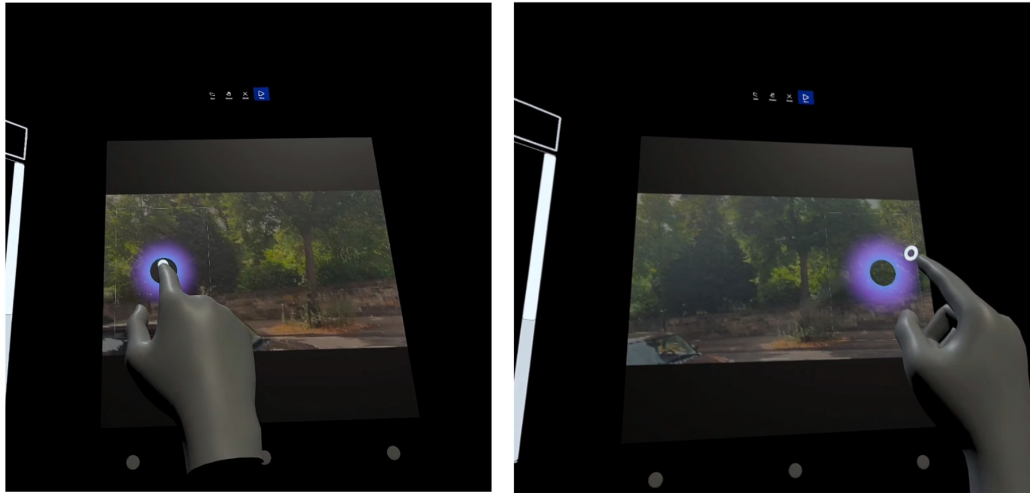


FIGURE 2.18: Video Player.

If the prefab is located on the wall or any other flat surface of a object, users would get physical feedback when pressing on the 3D screen, which is very similar to pressing the screen on smartphone or Ipad to pause or fast forward a Youtube video.

2.5 Painting Board

Writing or painting in virtual space is a topic that's been discussed for a long time-unlike drawing or writing on a 2D plane, define a range of writable place could be a difficult step. A lot of air-writing technology are based on complicated tracking method or hardware support, in order to achieve a higher level of freedom in using, this widget includes only OVRHand finger-writing.

Painting board widget enables users to leave message or hand-drawing anywhere in mixed reality environment, they could place the prefab at a desired position and add a temporary message in the space.

2.5.1 Prefab

The painting board prefab contains a simple black plate and UI panel, users could draw or write on the board with fingers, the usage is very direct and intuitive. It could be placed anywhere in the scene and used as a temporary notice board or graffiti board.

And by pressing the 'Clear' button on UI panel, all the drawings would be cleaned.

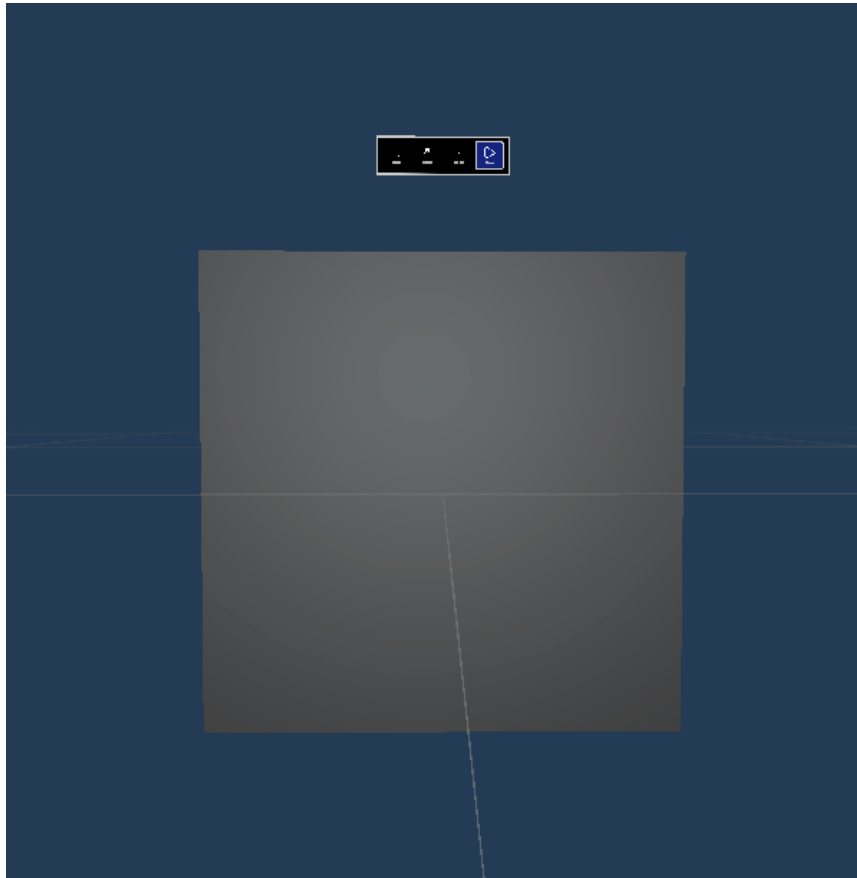


FIGURE 2.19: Painting Board Prefab.

2.5.2 Generating and Placement

The generating process of painting board prefab is very similar with those for screen prefab. When the “Generating painting board” checkbox on the control panel is switched on, users could do pitch and drag in the air to define the size and position of the painting board.

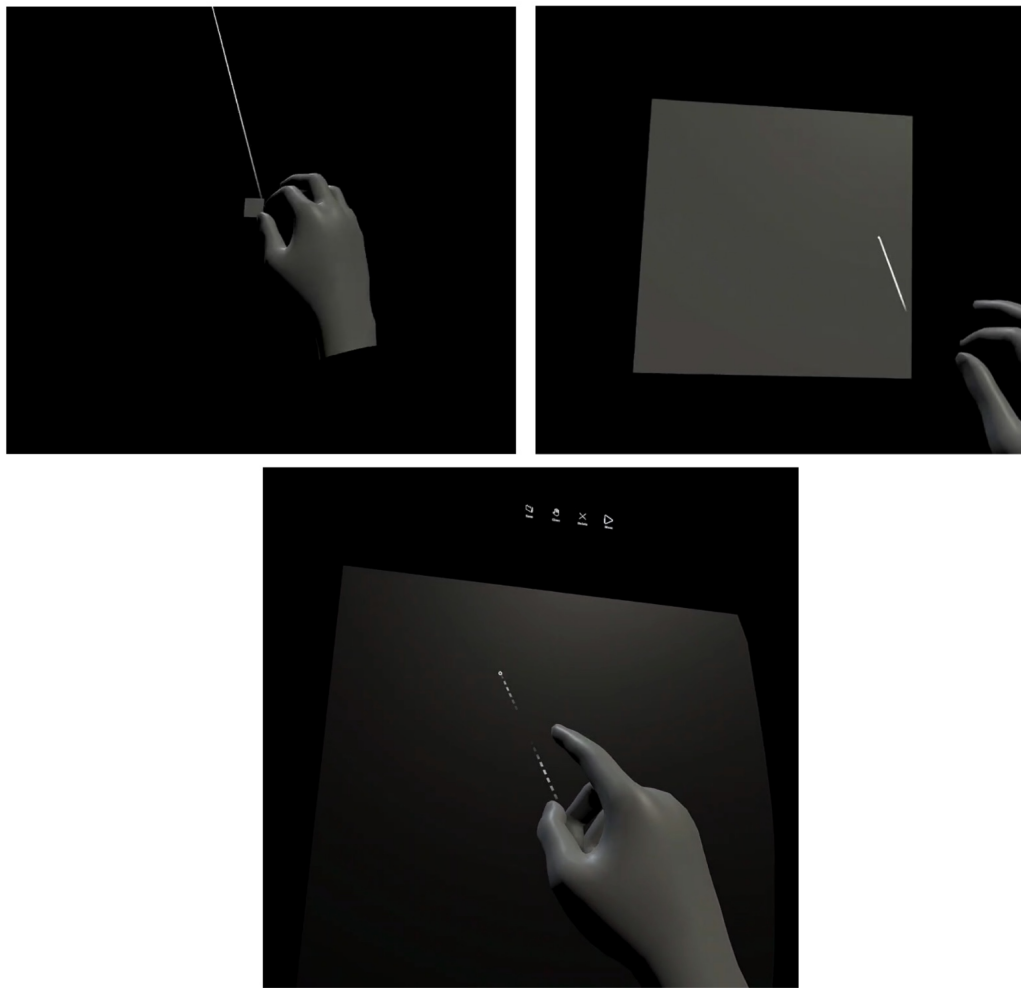


FIGURE 2.20: Painting Board Generating.

2.5.3 Gesture Control

For generating a default size writing panel, a more direct and visualized approach is introduced in this project. Gesture detection allows user to place a new painting board by a simple signal.

If the board prefab is placed against a flat surface, users could get haptic feedback during writing and painting. In order to provide a more accurate placement, this project offer a hand gesture generating method. After switching on the 'Gesture Mode' checkbox on control panel, users could generate a new painting board by hold their palm in the air for a certain period of time(several seconds), and the board prefab will be placed exactly at the position of their hand.

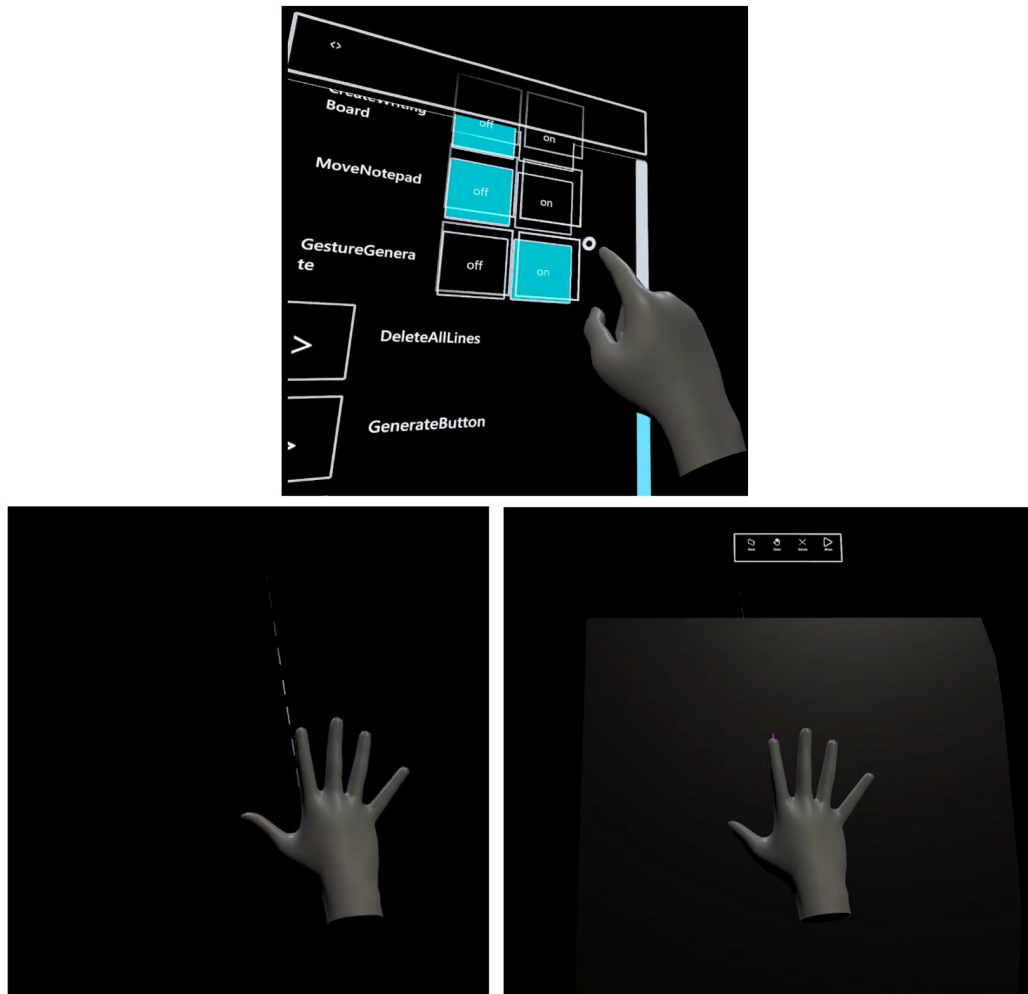


FIGURE 2.21: Gesture Control.

2.6 Notepad

Three-dimensional notepad widgets is a simple idea of involving the traditional notepad app in immersive environment.

2.6.1 Prefab

Notepad prefab includes a blank input box for word inputting and a user interface panel for additional functions. Since there is no keyboard hardware support in virtual environment, and even with pass-through, it will be hard to operate computer or laptop in real world, so a hidden virtual keyboard is included in the prefab, the virtual keyboard would be activated while inputting.

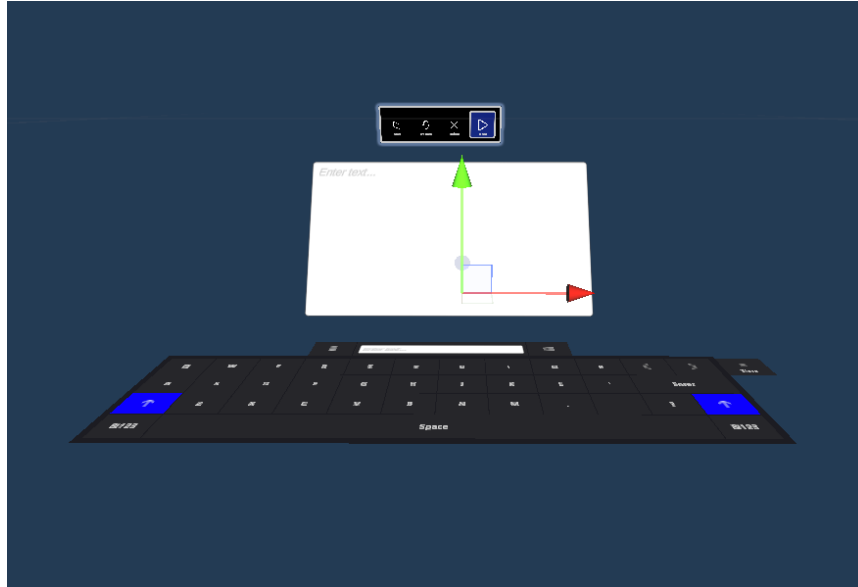


FIGURE 2.22: Notepad Prefab.

A lot of the information storage in immersive experience are inside the virtual environment, some of them could be erased after the end of experience, so further access of these information could be difficult, which might lead to low working efficiency.

By pressing the 'save' button on UI panel, the text in notepad written by users could be saved as a new file in the headset and exported to laptop if needed.

2.6.2 Generating and Placement

The generating and placement of Notepad prefab is similar to those of command button. Users could generate a new notepad by pressing on the control panel. And they could also attach the prefab to their left index fingertip when they pinching with right hand after turning on the 'Move Notepad' command on panel.

By aligning the keyboard to horizontal object surface like desks could enable users to achieve haptic feedback from accessible objects while typing.

Chapter 3

Authoring

3.1 Overall Concept

Authoring is part of the core of this thesis, this project provides users with the freedom to create, and define the operation system of scene objects according to their own preference.

For each user to experience the immersive scene, not only the 3d user interface settings are different, but also the operation method for the display items in the scene is different. Users are free to define how to operate the objects in the scene, and to get a personalized and creative immersive interactive experience.

In this project, every 3D widgets prefab and operable scene object come with authoring slots.

And multiple slots for the display objects representing different controllable attributes.

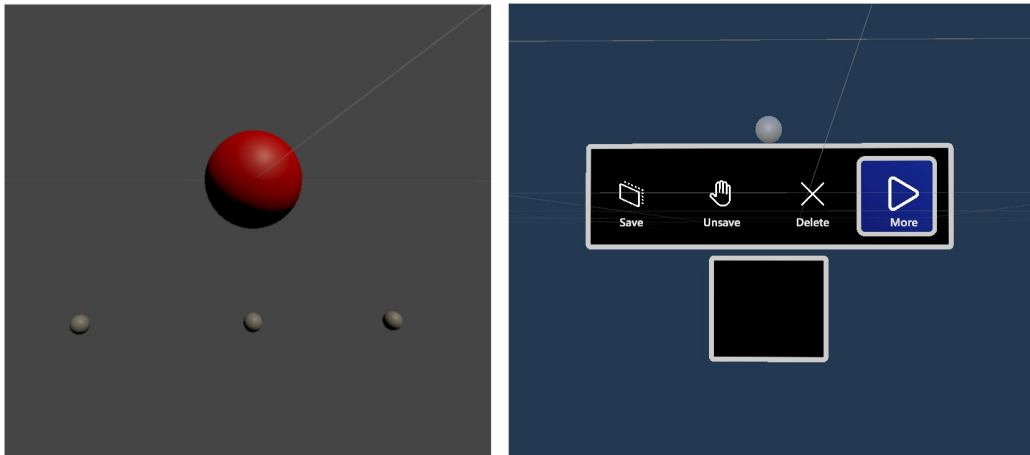


FIGURE 3.1: Scene Object Slots and Prefab Slots.

The project aims to enable users to link the slots on the 3D widgets to the slots on displayed objects by their virtual hands. By pinching near the start slots they

could generate a visible line on their fingertips, and complete the linking by releasing pinching near the end slots.

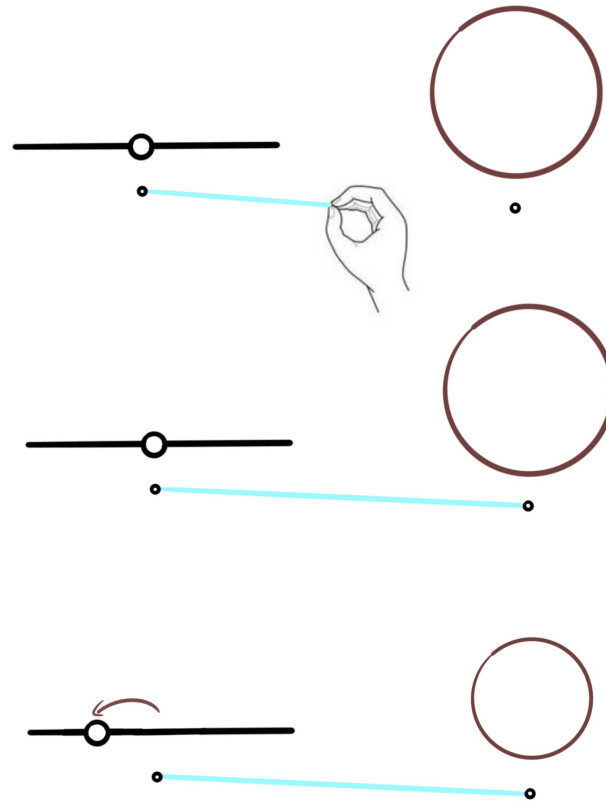


FIGURE 3.2: Authoring Design.

This system is designed to be intuitive and direct for the users, they could pair the command widgets and signal receiver with lines and clearly see which object is controlling by which widget.

3.2 Realization

The authoring mode could also be switched on and off by the control panel, which would not interpolate other operations. In authoring mode, the system will detect a nearest widget slot (within a minimum distance) and create an authoring line, which could be linked with a interactive object in the scene.

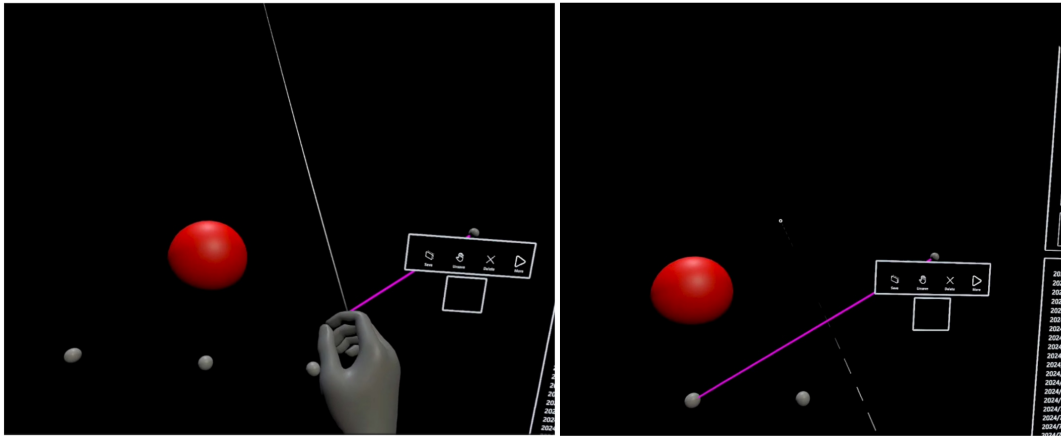


FIGURE 3.3: Authoring.

If there is an error in authoring, or users want to change the linked object, they could press the 'delete all lines' command on control panel and do authoring again.

3.2.1 Authoring with Scene Objects

Every slots on the scene objects represents different attributes of the objects, and the operating effect differs according to the type of the widgets it is linked with.

For example, changing size of the sample visualized object is discrete with command button, By linking button prefab to object slot and pressing would trigger a size change of the sample object, but without a accurate control of object size.

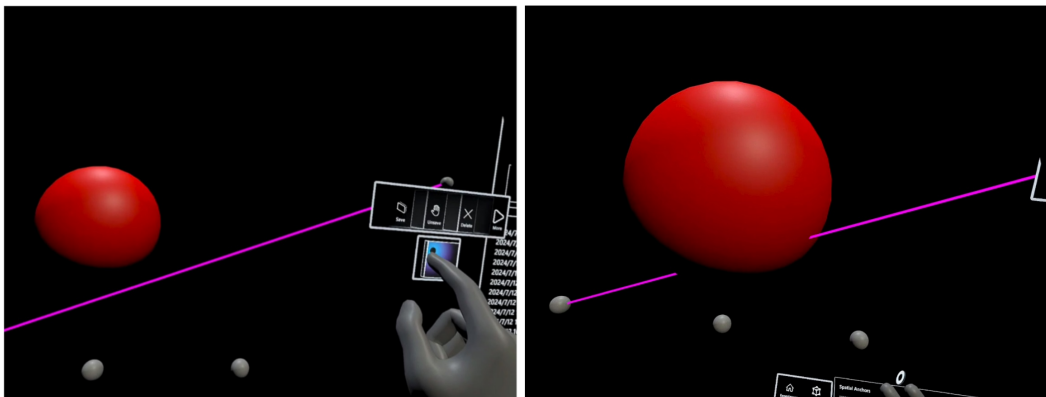


FIGURE 3.4: Button with ExampleVis.

On the other hand, adjusting the size of sample object is continuous with slider, and users are able to control the change and determine the size of the object by watching the value on slider prefab.

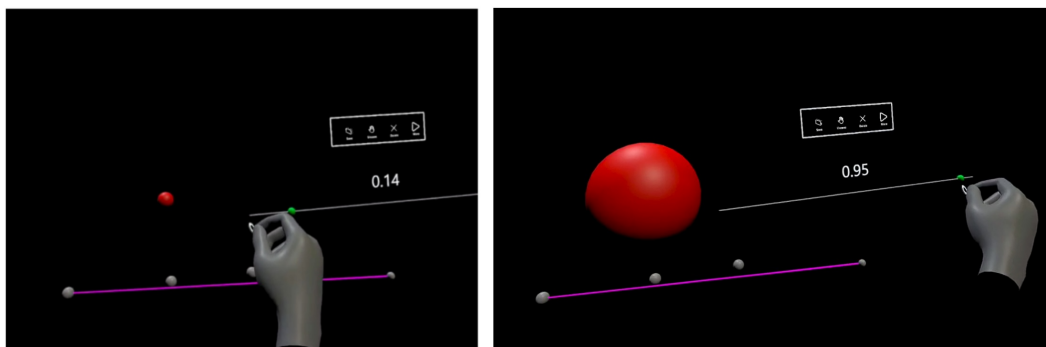


FIGURE 3.5: Slider with ExampleVis-1.

At the same time, another slot under sample visualized object contains the function of changing hue of the object. A slider track could cover a complete color wheel.

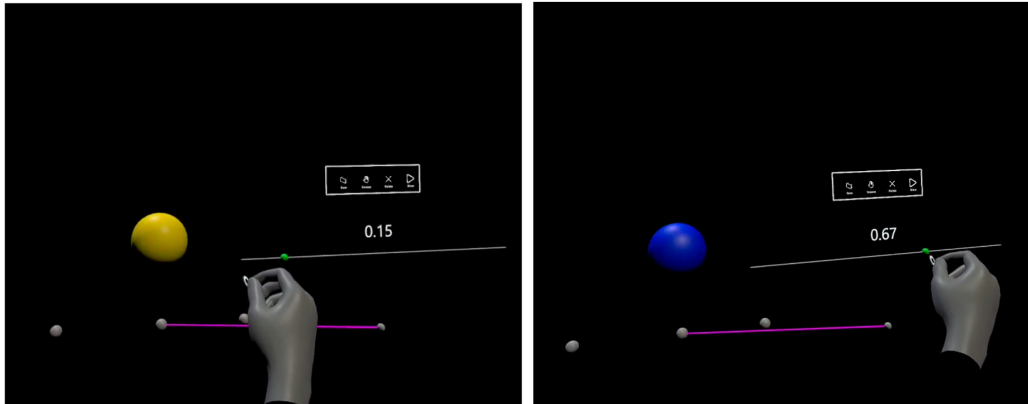


FIGURE 3.6: Slider with ExampleVis-2.

3.2.2 Authoring with Other Widgets

In addition to the actionable objects provided in the scene, the widgets can also connect to other widgets that support authoring.

When the screen prefab is activated as a image player, the two bottom slots could be linked with command buttons and provide image or slides flip-flop function.

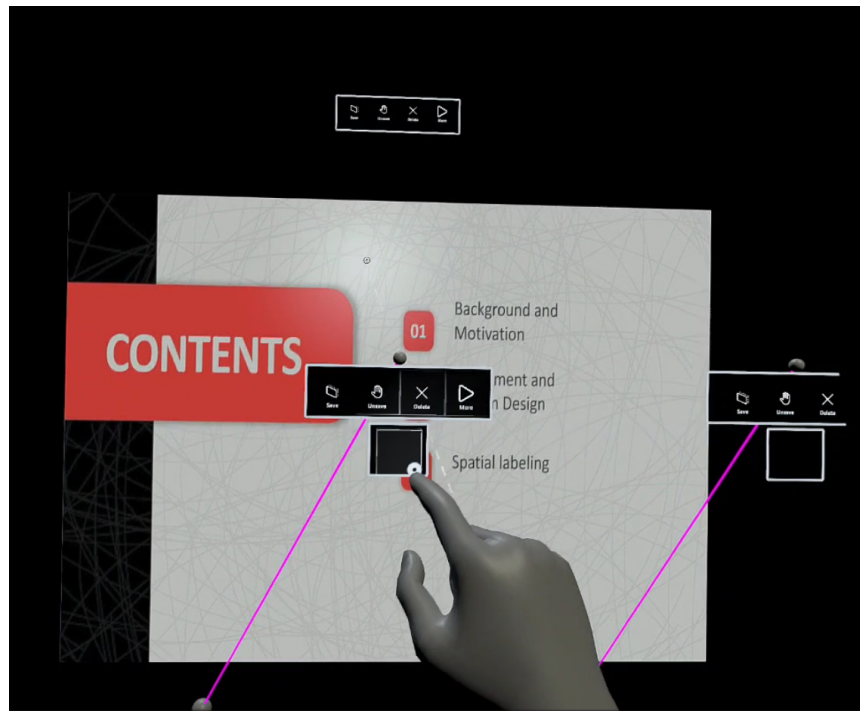


FIGURE 3.7: Button-Slides Flip-Flop.

When the screen prefab is activated as a video player, the three bottom slots could be linked with command buttons and provide video pause and start, video fast forward or backward function.



FIGURE 3.8: Button-Video Pause/Start/Forward/Backward.

Also, the slots could be connected with sliders and used that as a progress bar to control the video.



FIGURE 3.9: Slider-Video Progress Bar.

3.3 Authoring usage

Authoring enables users to freely define the control relationships between widgets and operable objects, which is the main idea of a opportunistic tangible widgets-based user customized interface.

This system breaks the traditional concept of "Objects could only be controlled by their own attached widgets", which not only means users could make their own decision to arrange the user interface composition according to their preference. Widgets do not need to be placed on or close to target object, but possible to operate remotely, which means that situations like widgets are spatially difficult to interact or some widgets might block the view would not appear during experience.

Multiple slots enable users to link more than one widget to the actionable target and adjust different attributes at the same time in the manner to which they are accustomed, this could bring a much more precise and convenient operation in mixed reality environment.

3.4 Further Possible Design

This project realized a primary concept of opportunistic 3D widgets authoring design, with more kinds of widgets prefabs and more sophisticated operable objects in the future, it needs to balance between more operating choices and convenience of usage.

Here lists some possible authoring design improvements:

3.4.1 Visualized Link Target

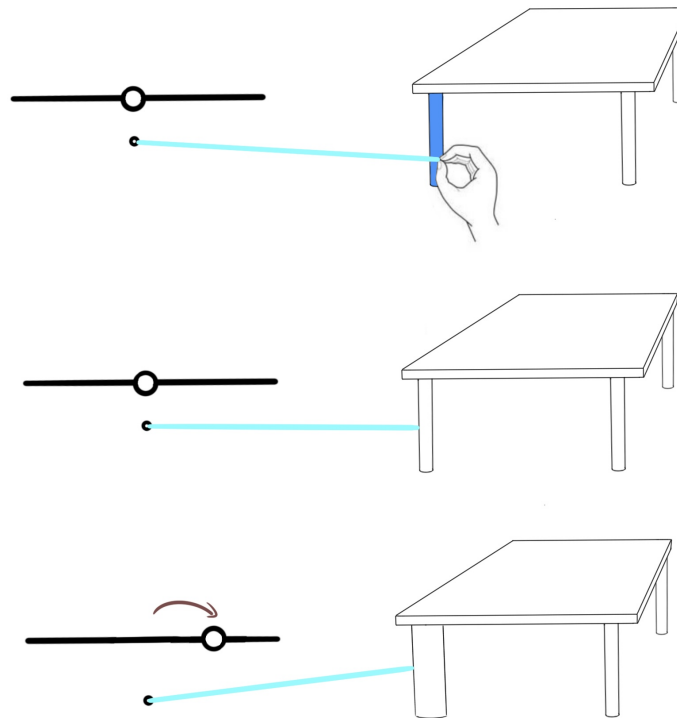


FIGURE 3.10: Visualized Link Target for Authoring.

When authoring with a object with multi deformable components, instead of link authoring line to default slots, users could link lines directly to a part of the object.

The object part would be highlighted in another colour when users' hand is close, to indicate that this deformable part of the object is targeted by authoring line. Releasing pinching here could complete authoring.

Operating widgets would trigger change in linked part, which is more clear and direct for users to define their operation.

3.4.2 Attribute Label

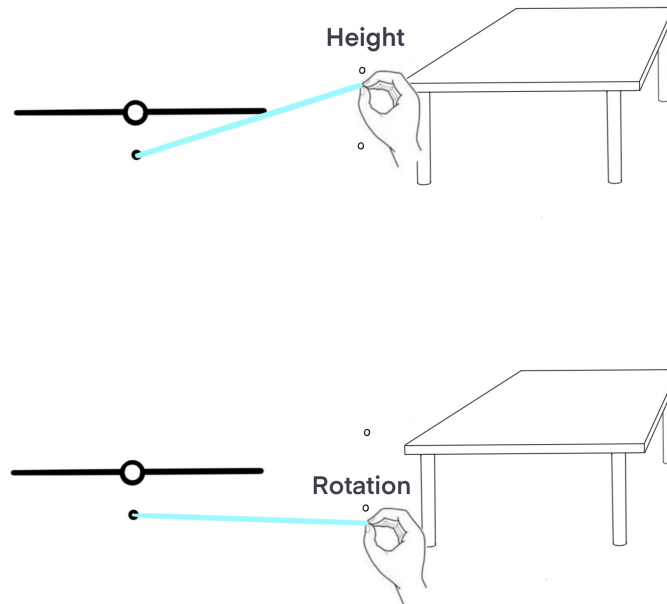


FIGURE 3.11: Attribute Label for Authoring.

Some of the object attributes like size, height or rotation are hard to be represented in a visualized way, so they could be shown by text.

The attribute this slot is controlling would be shown by labels when authoring line is close to it.

3.4.3 "Assembling"

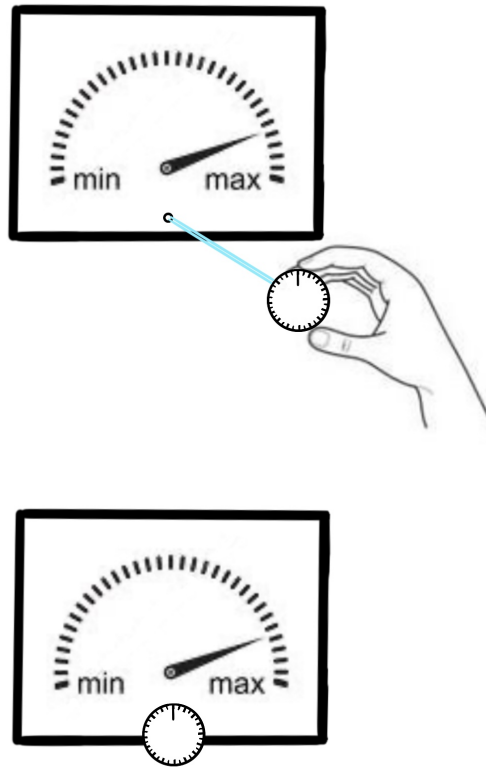


FIGURE 3.12: Assembling Function for Authoring.

Some widgets might be more suitable to be a part of the operable object according to common using habits. So authoring may also allow user to "assemble" widgets on other objects.

Widgets would be glued on target object if it is closed enough in authoring mode.

Chapter 4

Opportunistic User Interface with Spatial Labeling

4.1 Idea of Room Labeling

In a practical use for Mixed Reality, spatial alignment and labeling is an indispensable basic part of the scene. Marking the outline of objects could indicate edge of the event space and state interactive objects. The alignment of virtual scene and real environment would ensure users' safety and support interactive experience.

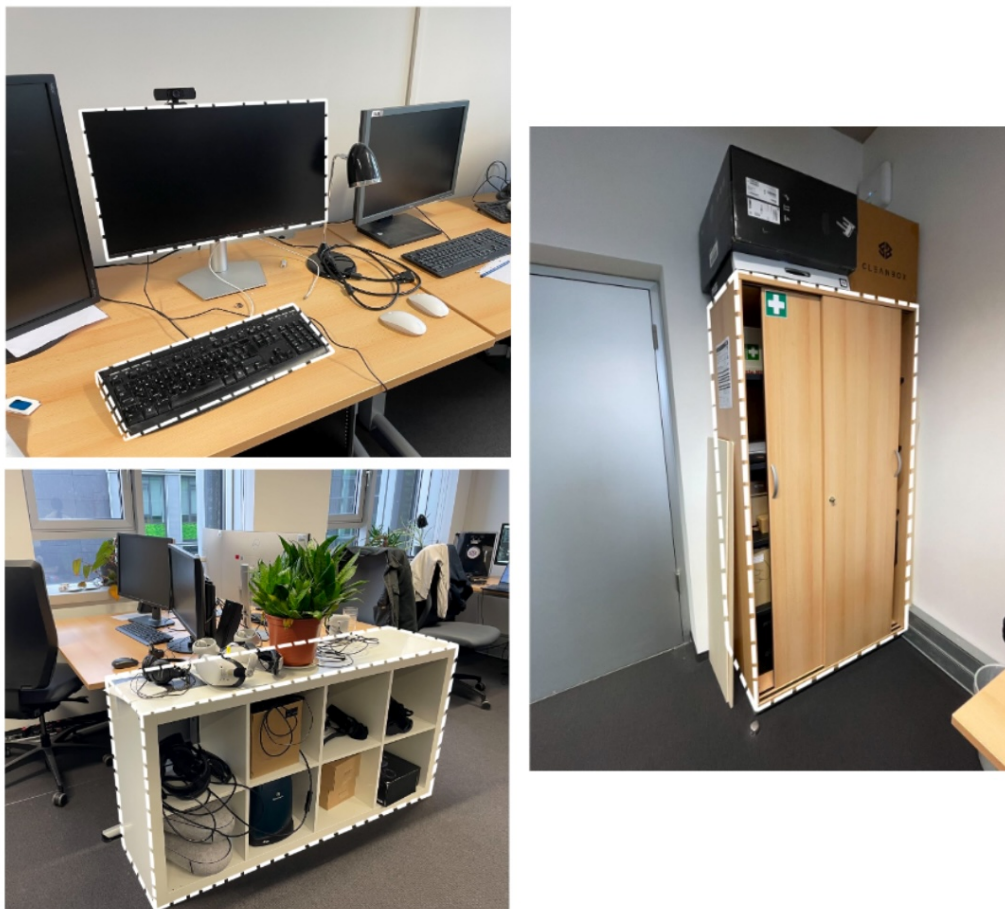


FIGURE 4.1: Room Labeling Design.

In order to implement the idea of less dependency on external hardware, the spatial spatial localization of the virtual scene is based on spatial anchors. This experiment assumes that the accessible space for users in a given scene has been spatially calibrated and labeled in advance, so that users can load all the demanded spatial anchors and 3D labeling

4.2 Introduction of 3D Labeling

The 3D Labeling for this experiment includes spatial anchor, AOI and SOI, which can help with spatial alignment, label scene objects, and also serve as a bridge for further development of the integration between opportunistic tangible widgets and mixed reality environments.

4.2.1 Spatial Anchor

Spatial anchor is provided by OCULUS, which helps users to add persistent objects to physical world. It is based on point cloud of the environment and enables cloud backup and sharing, which can be stored and persisted and later queried by the device that created it or any other supported device.

There would be inevitable error for a system based on pure software spatial alignment. A study about the stability and accuracy of spatial anchor has already been covered in my former work, as well as the possible improvement of a spatial anchor-based program.

4.2.2 SOI and AOI

In my former work of Spatial Alignment and Labeling for Immersive Content Organization, the concept of Area of interest and Sphere of interest were mentioned as 3D spatial labeling for large-scale immersive scene.

Area of interest is a cubic that defines an area of three-dimensional space, which could be used to present objects or obstacles need to be placed when the modeling of the scene is not yet complete or mark where user-defined objects can be placed.

Sphere of interest delineates a square flat area in which the user can place slides, images, and screen for display, as well as flat objects, qr codes, and other functional 2d objects.

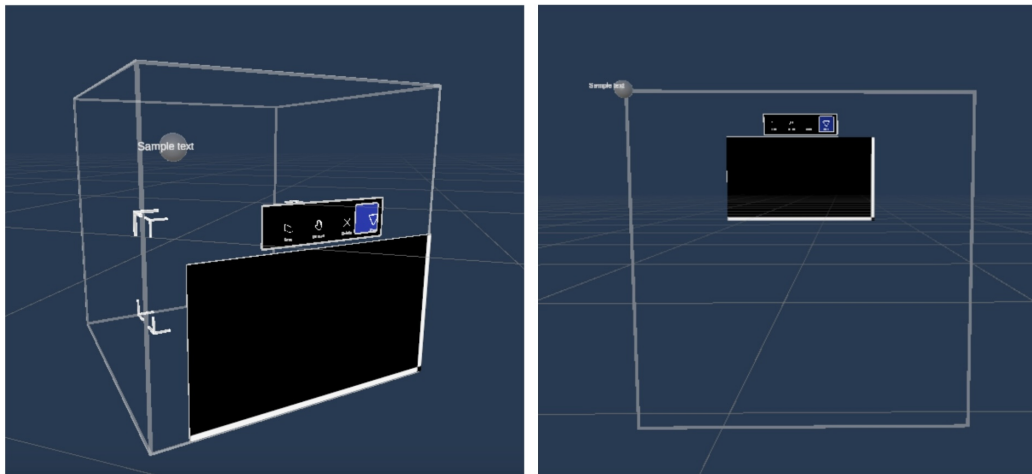


FIGURE 4.2: AOI and SOI.

In addition to the original usage for objects placement and position marking, now the labeling could also be used with 3D opportunistic widgets.

And the generating, saving and loading of AOI and SOI were already covered in my last project, so a completed predefined spatial labeling of a certain area, consist of spatial anchor and 3D labeling can be put into operation right away.

The labeling data could be created by the designer and stored in the headset, then uploaded to cloud and be accessible to other users.

4.3 Room Labeling Setup

Label data are already saved on cloud and in mixed reality headset, so after first calibration with calibration stand, users could load them directly. Furthermore, they could also calibrate by setting a spatial anchor to an appointed location as the original point and ensure the accuracy of room labeling alignment.

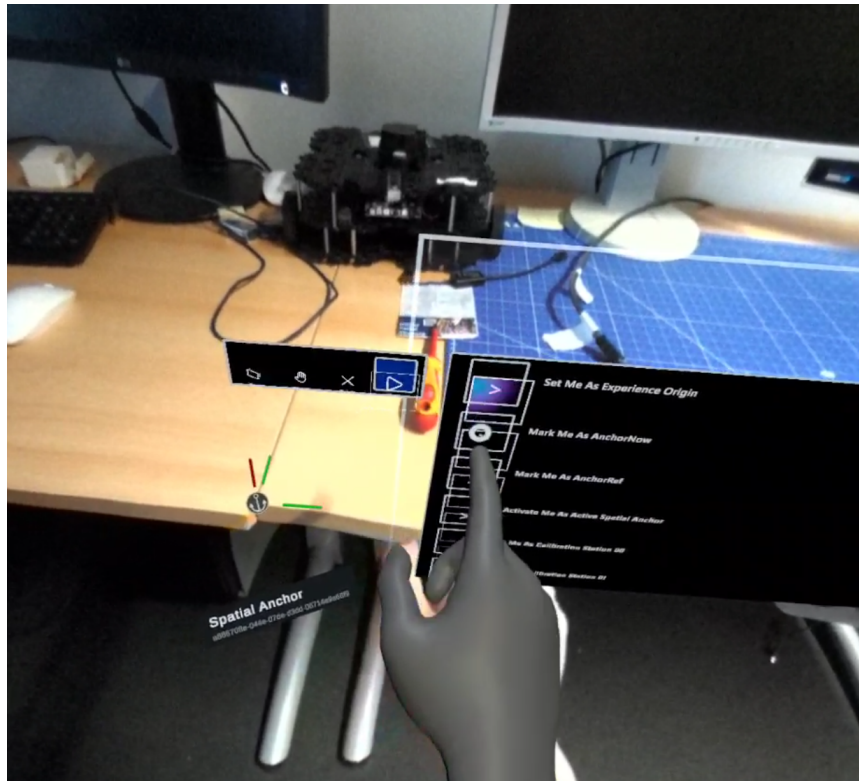


FIGURE 4.3: Set Spatial Anchor.

Then, pre-defined AOI and SOI system could be loaded by simply press "Load" button on hand menu.



FIGURE 4.4: Room Labeling.

4.4 3D Objects with Spatial Labeling

In a comprehensively labeled mixed reality room such as a laboratory or a exhibition hall, the silhouettes of scene objects would be outlined by spatial labeling and their position, center, size and surface would be clearly defined. So some of the opportunistic widgets placement could be easier with spatial labeling.

If the operation panel or information panels is moved near a spatial anchor or one side of the AOI, it could be attached to it automatically. Since the labeling are already delineating contours of flat objects (like desks, cabinet, bookshelf), panels attached to them would naturally get haptic feedback from surface of the objects.

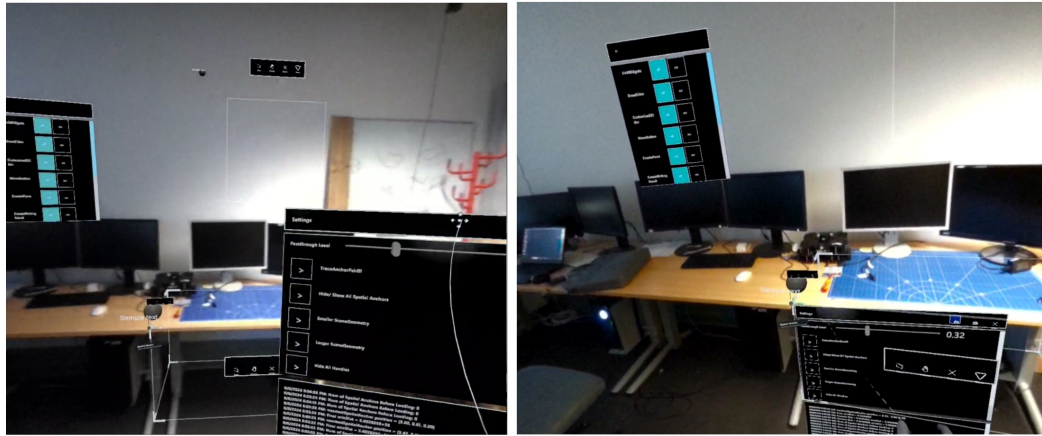


FIGURE 4.5: Panel Attached to Spatial Anchor/AOI.

If the 2D widgets like video player or image player is moved near a SOI, it could be snapped to its position and adjusted to its size automatically. Since the size and location of the labeling are pre-defined by the designer, in the vast majority of cases they are aligned to get physical feedback from object surface and designed to be a suitable size for the immersive scene. By automatic snapping users don't need to change the widgets condition again by themselves.

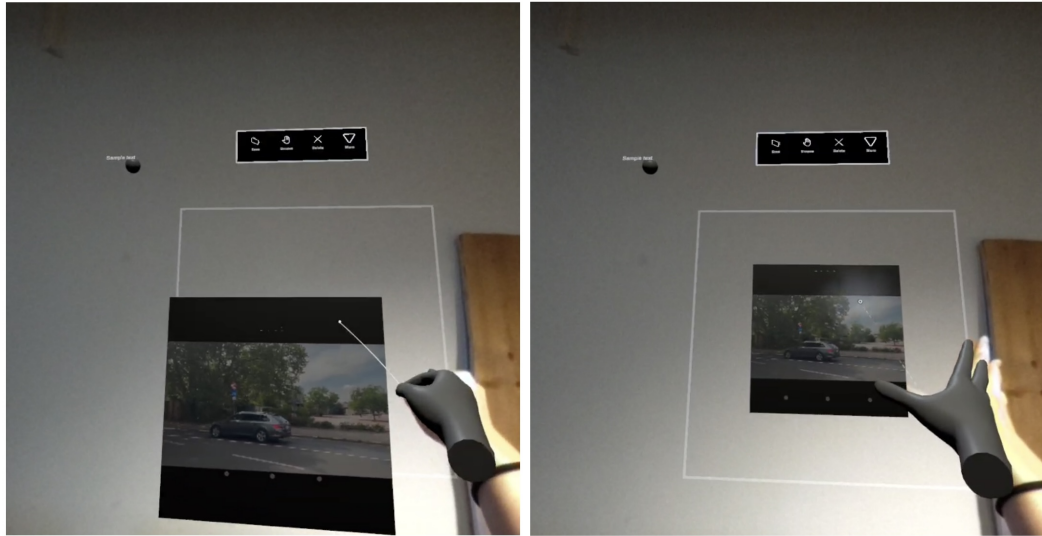


FIGURE 4.6: Video Player Attached to SOL.

Chapter 5

Discussion

5.1 3D Widgets in Unlabeled Space

Last chapter has covered the performance of 3D opportunistic widgets in spatial labeled spaces, and the approaches for the spatial labeling to help with the alignment of 3D widgets. However, beyond that, the design of opportunistic widgets is also suitable in non-labeled environment in many aspects.

With the help of pass-through function of mixed reality headset, users could have a unobstructed view of the surroundings, which in fact solved the conflict between user movement and limited movement space in traditional virtual environment systems. Users are now able to walk outside the restricted room without safety issues. And a large part of the opportunistic widgets are actually designed for unlabeled outdoor conditions.

Since the default spatial localization of the virtual objects are based on cloud point technology, which means based on the information of surroundings, it will not move along the activities of the users, preconfigured user interface could meet the problem of being hard to access when the users have walked away from their calibration point. A instant generation user interface consists with opportunistic widgets could be created anywhere according to users' demand and be moved to a ideal position anytime.

On the other hand, a toggle-along preconfigured user interface might block the view of users, which could lead to difficulties in utilization or even security problem. On the contrary, a user interface generated and arranged by themselves would eliminate the problem completely. Also, a lot of physical contact-based generating and placement approaches are involved in the 3D widgets design, anything in our daily environment could be used as a medium for haptic placement and then get physical feedback from.

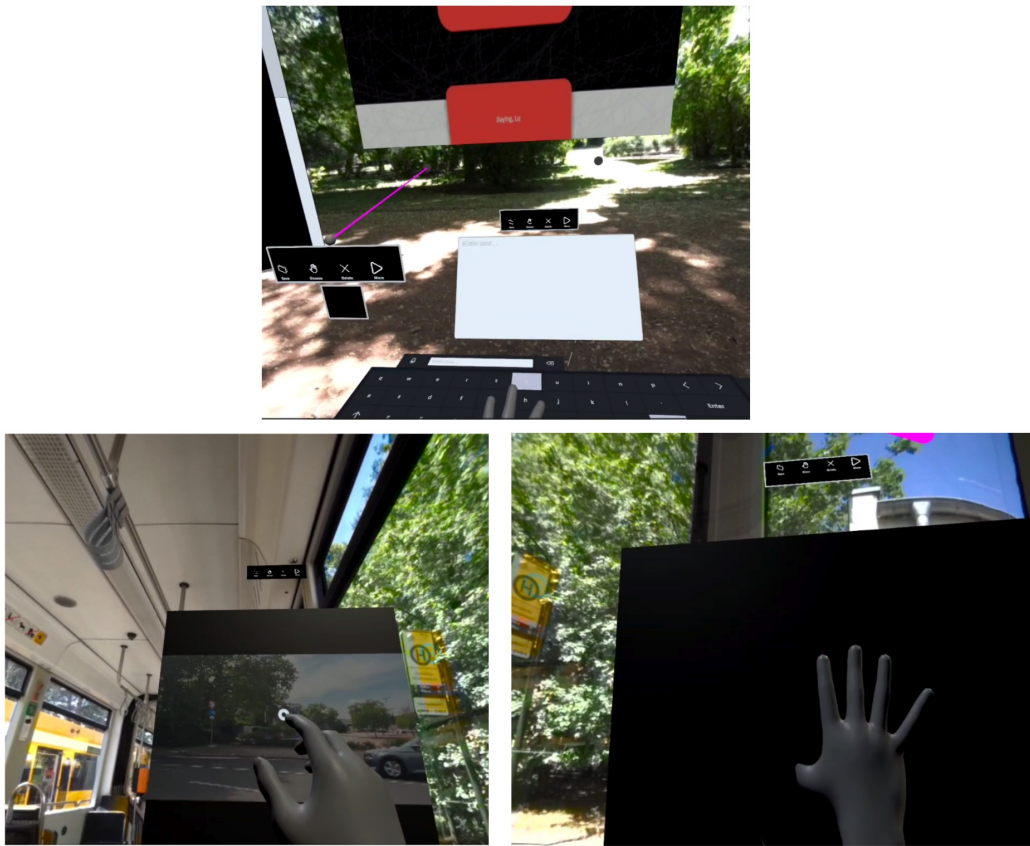


FIGURE 5.1: Non-labeled Environment Physical Feedback.

Easily using real world objects to help the mixed reality experience is a critical part of this project, turning surroundings into a part of the immersive experience could significantly widens the application scope of the system

5.2 Additional Functions and Remaining Problems

5.2.1 Specific or General Widgets

This project mainly focus on general 3D opportunistic widget prefabs, which is mostly fundamental component of a virtual user interface system, and the generating and placement with physical feedback do not have a strict requirement for target real world objects. Almost any kind of objects with touchable surface could provide demanded haptic feedback for users.

On one hand, general widgets have very little limit while using, and could easily apply to both labeled and unlabeled surroundings. However, specific widgets prefab designed for certain real world objects may be able to bring better feedback experience, if the system is primarily for a specific space with plenty of certain usable objects, a special designed prefab could perform better than general widgets.

In future design and practice, a combination of general and specific widgets could be a direction of development.

5.2.2 Voice Command

Oculus package has provided voice SDK functions, it enables users to bring voice interactions to app experiences and made voice command possible in the immersive systems. Voice command could shortcut some operations and make system more natural and flexible to interact with.

A large feature of the voice SDK is that it does not only support a simple word trigger to function logic, but based on artificial intelligence recognizing functionality. A configured “generate a button” command could also be trigger by sentences with similar meanings like “create a button” or “make a new button”, with this feature the users don’t need to mechanically memorize all the settled commands, instinctive speaking for their demand has a good possibility to reach their mission.

5.2.3 Better User Guiding

The design for operation of this system is aimed to be intuitive and direct, but some guiding for the users are still necessary during usage.

The project need to find a balance between maximizing the haptic feedback from surroundings in mixed reality environment and making the guiding clear and concise. The system should be able to operate even for first-time user without additional help.

Simple word notice or instruction could make a clear guiding, but too much reading in a virtual environment might reduces the comfort level of the usage experience or even cause dizziness. Some animated demonstration or other kind of instruction in virtual scene might be better approach.

5.2.4 Machine Learning and Real-Time Image Processing

This program is dedicated to less dependent on additional hardware support, the problems of spatial alignment and calibration has already been solved, but tracking a moving object without additional support like QR Code or external camera could be very difficult.

By now users could only get haptic feedback from stable real world objects, if an object carrying the alignment of a widget prefab has been moved, users need to place the virtual interface again to get haptic interaction.

Attaching functional widgets on moving substance is not yet supported in the system, so some actions like using a stick as the simulation of holding a pen to write in immersive experience id not possible right now. If the idea of less dependent om external hardware is insisted to stay, the combination of real-time view from headset camera and the help of machine learning and real-time image processing may be able to provide demanded assistance.

Meta has introduced SAM2 as segment model, which supports real-time promptable object segmentation in images and videos, so alignment with real-time segmentation could be a way forward.

5.2.5 Travel Mode

This year in May, Meta has announced Travel Mode on Meta Quest 2 and 3 headset, which lets users use the headsets on moving transportation like cars or flights (as a passenger). Systems working in a moving situation could be a potential.

The release of Travel Mode is later than the start of this project, so it is not yet considered in the system. Opportunistic 3D widgets in this project could be moved

by the users, but it still does not apply to high-speed moving transportation. Also, the spatial anchor system is based on cloud point and environment information, which would not be suitable for usage in motion.

Chapter 6

Conclusion

6.1 Summary

This thesis focuses on the possibility of users turning surroundings into interactive functional 3D widgets and get haptic feedback for a better mixed reality experience. So the project was conveyed by three main parts: Opportunistic tangible widgets prefabs, Authoring and real usage of customized interfaces in labeled and non-labeled environment.

Opportunistic widgets prefabs in this project consists three-dimensional common 2D widgets and practical control tools in mixed reality. The thesis realized some of the existed concept and raised further designs.

Authoring allows users to creative arrangement for control target of opportunistic widgets and adjust interface by their own preference, which provides users with more operation freedom and could adapt to different situation.

This project realized interaction between widgets and spatial labeling with achievement of a previous research project, and has tested real practice of the system in both labeled and non-labeled environment.

This thesis aims to create possibilities for users to construct a pure software opportunistic interface with hand tracking. Even with several remained problems, this thesis has solid outcomes and shows feasibility and potential of opportunistic widgets in immersive media.

6.2 Conclusion

As demonstrated in the thesis, opportunistic interface system in mixed reality has a huge potential. The industry is still a vast field to be explored and there is a lot of directions and topics to work on. But it also means that the possibilities are endless, and with time and creativity we can realize almost anything we have imagined.

With limited time and hardware support, this project shows the possibility of mixed reality headset goes out of a limit space and become a part of other activities, which could be a first step for immersive media to walk into our daily life

Appendix A

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